higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ENGINEERING SCIENCE N4

3 APRIL 2019

This marking guideline consists of 14 pages.

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✓= 1 mark
√ = ½ mark

QUESTION 1

1.1 1.1.1 When pressure is exerted in an enclosed system it will be transmitted with the same intensity throughout in all directions.

1.1.2
- When you use the same force to push a truck and a car, the car will have more acceleration than the truck, because the car has less mass.
- It is easier to push an empty shopping cart than a full one, because the full shopping cart has more mass than the empty one, therefore more force is required to push the full shopping cart.

(2 × 2) (4)

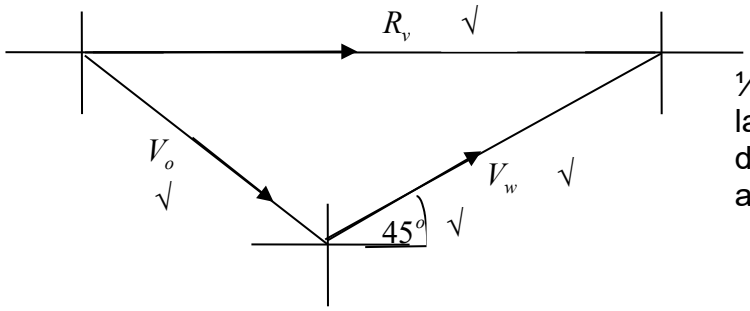
1.2 1.2.1 Deformation of a material due to internal stress (2)

1.2.2 Sum of all forces to the left or right of a section on a beam. (1)

1.3
- Air or gas accumulator
- Spring accumulator
- Mass accumulator

(Any 2 × 1) (2)

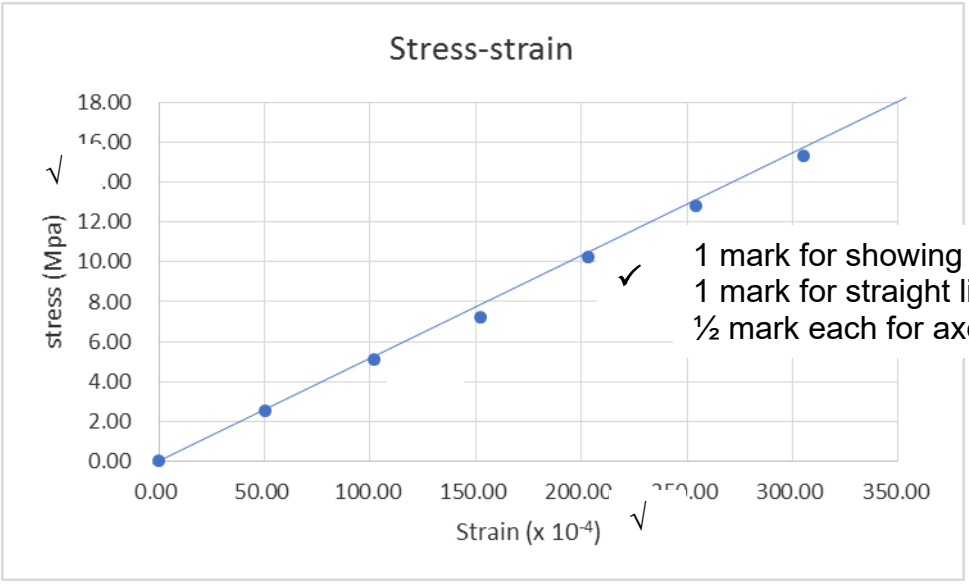
1.4 Triangular velocity vector diagram.



½ mark for each correctly labelled side with correct direction as well as correct angle

(2)

1.5



1 mark for showing the dots
1 mark for straight line
½ mark each for axes

(3)

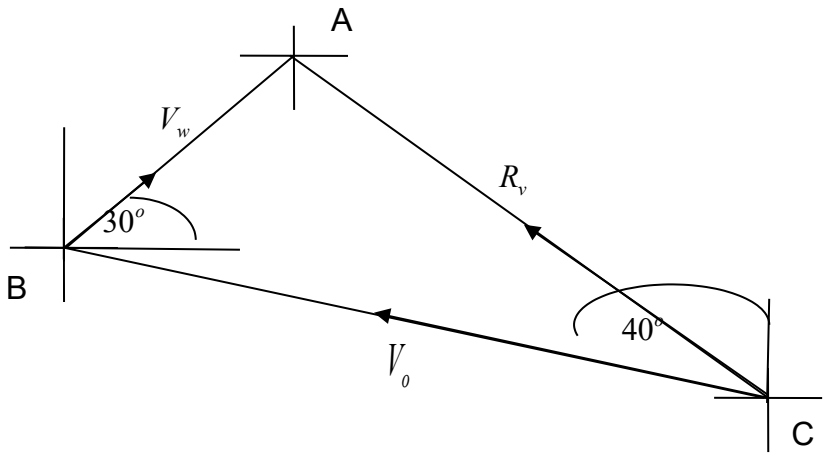
[14]

QUESTION 2

2.1 $V_o = 360\ km/h\ \ N40^o W$

$V_w = 120\ km/h\ \ W30^o S$

$R_v = ...$



$B = (30^\circ + 50^\circ)$

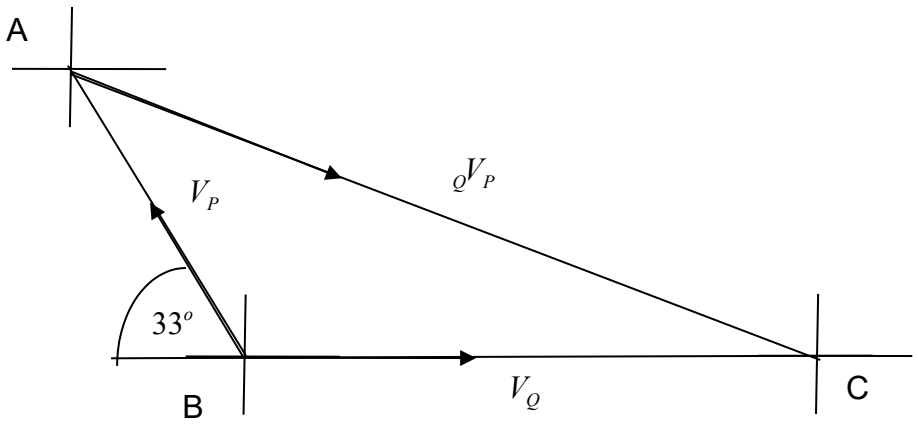
$= 80^\circ$ ✓

$$b = \sqrt{a^2 + c^2 - 2acCosB}$$
$$b = \sqrt{360^2 + 120^2 - 2(360)(120)Cos80}\ \checkmark$$
$$= 359{,}161\ km/h$$
$\checkmark$

$$\frac{SinC}{C} = \frac{SinB}{b}$$
$$C = \sin^{-1}(\frac{210,Sin80}{359{,}161})\ \checkmark$$

$$= 24.56^\circ\ \checkmark$$
$$\theta = 40^\circ - 24.56^\circ$$
$$= 15{,}44^\circ$$
$$R_v = 359{,}161\ km/h \quad N15{,}444^\circ W$$ (5)

2.2 $V_P = 270\ km/h\ \ W33^\circ N$
$V_Q = 220\ km/h$ E
${}_QV_P = ...$



$$B = (57^o + 90^0)$$
$$= 147^0\ \checkmark$$

$$b = \sqrt{a^2 + c^2 - 2acCosB}$$
$$b = \sqrt{220^2 + 270^2 - 2(220)(270)Cos147}\ \checkmark$$
$$= 470{,}036\ km/h$$
$$\frac{SinA}{a} = \frac{SinB}{b}$$
$$a = \sin^{-1}(\frac{220.Sin147}{470{,}03})\ \checkmark$$
$$= 14.769^0\ \checkmark$$
$${}_QV_P = 470{,}036\ km/h \quad E18.231^\circ S$$ (4)

2.3 $S_h = 220\ m$
$u = 170\ m/s$

$t_{tot} = 2{,}1\ s$
$g = -9{,}8\ m/s^2$

$$S_h = u.t.Sin\theta + \frac{g}{2}t^2$$
$$u.t.Sin\theta = S_h - \frac{g}{2}t^2$$
$$= 220m - (\frac{-9{,}8m/s^2}{2}).(2.1)^2 \checkmark$$
$$= 241.609m$$
$$\sin\theta = \frac{241.609}{(180)(2.1)} \checkmark$$
$$= 0{,}639177 \checkmark$$
$$\theta = Sin^{-1}(0{,}639177248)$$
$$= 39{,}73^\circ \checkmark$$

(4)
[13]

QUESTION 3

3.1 3.1.1 B
3.1.2 C
(2 × 2) (4)

3.2 3.2.1 B (2)
3.2.2 C (2)
3.2.3 A (1)
[9]

QUESTION 4

$\theta = 28^\circ$
$m = 1\,200\ kg$
$S_{slope} = 15\ m$
$S_{horizontal} = 15\ m$

4.1 $E_{p(top)}$

$$E_{p(top)} = mgh$$
$$= (1\,200\ kg).(9{,}8m/s^2).15\ m.Sin28 \checkmark$$
$$= 82814{,}784\ J \checkmark$$

(2)

4.2 $E_{k(top)}$, 8m

$$E_{p(7m)} = m.g.8m.Sin\theta$$
$$= 1\,200\ kg(9{,}8\ m/s^2).7m.Sin28$$

$= 28646{,}899\ J$ ✓

$E_{p(lost)} = E_{p(top)} - E_{p(8m)}$

$= 82814.784J - 28646.899J$

$= 44167{,}885J$ ✓

$\therefore\ E_{k(8m)} = E_{p(lost)}$

$= 44167{,}885J$ ✓

$$v = \frac{\sqrt{2(44167{,}885}}{1\,200}$$

$= 8{,}58\ m/s$ ✓ (4)

4.3 a , v = o , 20 m

$E_{k(bot)} = 82814{,}784\ J$ ✓

$$v = \frac{\sqrt{2(82814.7840}}{1200}$$

$= 11.748\ m/s$ ✓

$v^2 = u^2 + 2a.S$

$$a = \frac{v^2 - u^2}{2.S}$$

$$= \frac{0^2 - (11.748\ m/s)^2}{2(20\ m)}$$

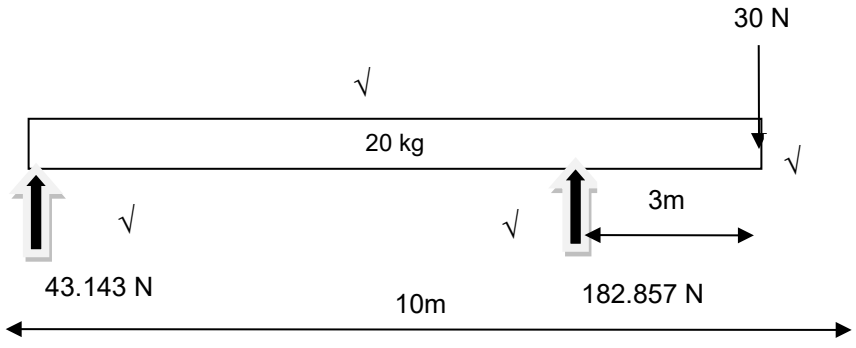
$= -3{,}450\ m/s^2$ ✓ (3)

4.4 $E_{k(20m)} \Rightarrow \frac{1}{2}.1\,200\ kg(0) = 0$ ✓ (1)

[10]

QUESTION 5

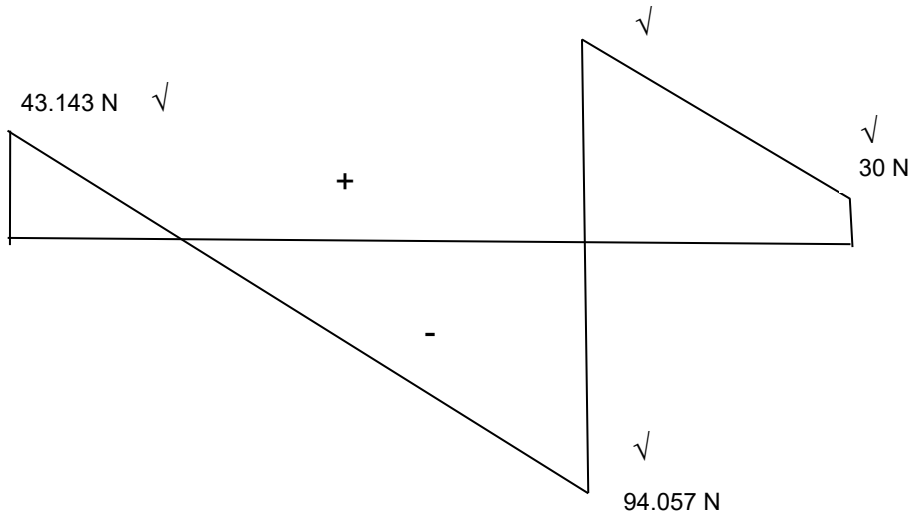
5.1 5.1.1



(2)

5.1.2

86.8 N



(2)

5.1.3

$$\sum F_D = 0$$

$$\sum F_D \Rightarrow 43.143N - 19.6N/m = 0$$

$$x = \frac{-43.143N}{-19.6N/m}$$

$$= 2.201m \checkmark$$

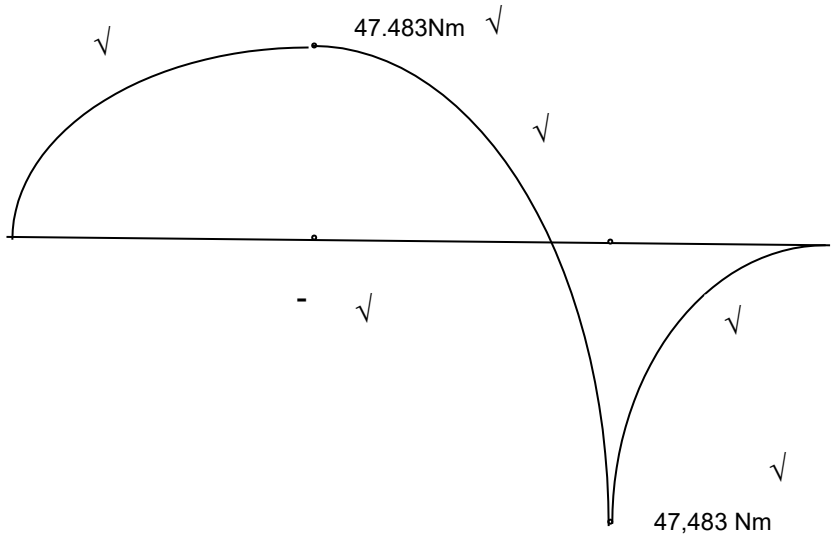
$$BM_E \Rightarrow 43{,}143\ N(2{,}201\ m) - 19.6\ N/m(2{,}201\ m)\frac{2.201}{2} = 47{,}483\ Nm$$

✓

$$BM_D \Rightarrow 43{,}143\ N(7m) - 19{,}6\ N/m(7\ m)\frac{7}{2} = -178{,}199\ Nm \checkmark$$

(3)

5.1.4



(3)

5.2

Shape	Volume	G	VG

(m^3)			
	$= \frac{\pi d^2}{4} S_l$ $= \frac{\pi(100)^2}{4}.(80)$ $= 628318,857$ √	$= \frac{h}{2}$ $= \frac{80}{2}$ $= 40$ √	$= (628318,857)40$ $= 25132754,28$ √
	$= \frac{\pi d^2}{4} S_l$ $= \frac{\pi(60)^2}{4}.(100)$ $= 282743,339$ √	$= h_1 + \frac{h}{2}$ $= 80 + \frac{100}{2}$ $= 130$ √	$= (282743.339),130$ $= 36756634,07$ √
	$\sum V = 628318,857 + 282743,339$ $= 911062,196$ √		$\sum VG = 25132754,28 + 36756634,07$ $= 61889375,28$ √
	$\bar{Y} = \frac{\sum VG}{\sum V} = \frac{61889375}{911062,196}$ √ $= 67,931\ mm$ √		

(5)
[15]

QUESTION 6:

6.1 $f_{eff} = 310\ N$

$S_l = 120\ mm$

$$d = \frac{1}{3}D$$

$MA = 20$

6.1.1 W_R , $\eta = 93\%$

$$\frac{W_R}{f_p} = \frac{D^2}{d^2} \qquad MA = \frac{f_p}{f_h}.\frac{100}{\eta}$$

$$f_p = \frac{f_h MA.\eta}{100}$$
$$= \frac{310N.(20).(93)}{100} \checkmark$$
$$= 5766\ N \checkmark$$
$$W_R = \frac{D^2.f_p}{d^2}$$
$$= \frac{(D)^2.(5766N)}{(\frac{1}{3}D)^2} \checkmark\checkmark$$

$$= 51894\ N \checkmark$$ (5)

6.1.2 n , $S_L = 480\ mm$ $\eta = 93\%$

$$V_a = \frac{\pi.D^2}{4}.S_L$$
$$= \frac{\pi(0.29m)^2}{4}.(0{,}41\ m)$$
$$= 0.027081314\ m^3\ \checkmark$$
$$V_S = V_a \frac{100}{\eta}$$
$$= 0.027081314\ m^3 \frac{100}{93}$$
$$= 0.029119692\ m^3\ \checkmark$$
$$\therefore\ \frac{\pi.d^2.S_l.n}{4} = 0{,}029119692\ m^3$$
$$\frac{\pi.(0{,}11\ m)^2.(0{,}12\ m)n}{4} = 0{,}038541326\ m^3$$
$$\therefore n = \frac{0.029119692\ m^3}{0.001140398\ m^3}$$
$$= 25{,}535 \Rightarrow 26 \checkmark \quad (3)$$

6.2 $n = 1$
$S_l = ...$
$d = 210\ mm$
$V_a = 0{,}042\ m^3$

6.2.1 S_l

$V_s = V_a$ since efficiency s 100%
$= 0{,}042\ m^3$ √
$$V_s = \frac{\pi.d^2.}{4} S_l n$$
$$= \frac{\pi(0{,}21\ m)^2.(S_l).1}{4} = 0{,}042\ r \ \sqrt{}$$

$$S_l = \frac{4(0{,}042\ m^3)}{\pi.(0{,}21\ m)^2}\ \checkmark$$
$$= 1{,}213\ m\ \checkmark$$

$$W = f_h.(S_l)$$
$$= 7500N.(1.213m)$$
$$= 9094.568\ J\ \checkmark \quad (4)$$

6.3 $W = 250{,}248\ kJ$
$d = 20\ m$
$P = 860\ kPa$
h

$$a = \frac{\pi}{4}d^2 = \frac{\pi}{4}(20\ m)^2$$
$$= 314{,}159\ m^2 \checkmark$$

$$W = P.V$$
$$V = \frac{W}{P}$$

$$V = \frac{250248\ J}{380\ P_a}$$
$$= 658{,}54737\ m^3 \checkmark$$
$$V = a.h$$
$$h = \frac{V}{a}$$
$$h = \frac{658{,}5737\ m^3}{314{,}159\ m^2}$$
$$= 2{,}096\ m \checkmark$$

(3)
[15]

QUESTION 7

7.1 $S = 40mm$ $d_f = 30mm$
$l_o = 800mm$ $l_o = 520mm$
$E = 116GPa$
$\delta_{round} = 380MPa$

7.1.1
$$\delta_{square} = \frac{F}{a}$$
$$F = \delta.a$$
$$= 380\times10^6\ Pa(\frac{\pi d^2}{4})$$
$$= 380\times10^6\ Pa(\frac{\pi(0{,}3\ m)^2}{4}) \checkmark$$
$$= 268606{,}1719\ N \checkmark$$

Square

$$\delta_{square} = \frac{268606{,}1719\ N}{(0{,}04\ m)^2}$$
$$= 167878750\ Pa \checkmark$$

(3)

7.1.2
$$\Delta l_{round} = \frac{F..l}{A.E}$$

(3)

$$= \frac{268606\ N.(0{,}52\ m)}{\frac{\pi(0{,}03\ m)^2}{4}.116\times10^9}$$
$$= 1{,}70345\times10^{-3}\ m\ \checkmark$$

$$\Delta l_{square} = \frac{F..l}{A.E}$$
$$= \frac{268606\ N.(0{,}8\ m)}{\frac{\pi(0{,}04\ m)^2}{4}.116\times10^9}$$
$$= 1.1578\times10^{-3}\ m\ \checkmark$$

$$\Delta l_{tot} = \Delta l_{square} + \Delta l_{round}$$
$$= 1{,}70345\times10^{-3}\ mm + 1{,}15778\times10^{-3}\ mm$$

$$= 2{,}862\times10^{-3}\ m\ \checkmark$$

7.2 $d = 35\ mm$
$l_o = 650\ mm$
$F = 80\ kN$
$\Delta l = 0{,}311\ m$

7.2.1
$$\delta = \frac{F}{a}$$
$$= \frac{80\ 000\ N}{\frac{\pi(0{,}035\ m)^2}{4}}\ \checkmark$$
$$= 83150337{,}62\ Pa$$
$$= 83{,}150\ MPa\ \checkmark$$

7.2.2
$$\varepsilon = \frac{\Delta l}{l}$$
$$= \frac{0.311mm}{650mm}\ \checkmark$$
$$= 4.78461\times10^{-4}\ \checkmark$$

(2 × 2) (4)
[10]

QUESTION 8

8.1 $l = 3\ m$
$t = 25\ ^oC$

$\Delta t = 380\ K$

$l_2 = 3{,}024\ m$

8.1.1 Δl

$$\Delta l = l_o.\alpha.\Delta t$$

$$\alpha = \frac{\Delta l}{l_o.\Delta t}$$

$$= \frac{0{,}024\ m}{3m.(380\ ^oC)}\ \checkmark$$

$$= 21{,}05 \times 10^{-6}\ /^o\ C\ \checkmark \quad (2)$$

8.1.2 Aluminum (1)

8.2 $m_1 = 12\ kg$

$P_1 = 170\ kPa$ $\qquad P_2 = 310\ kPa$

$t_1 = 21\ ^oC$ $\qquad t_2 = 38\ ^oC$

$R = 287\ J/kgK$

$T_1 \Rightarrow 21\ ^oC + 273 = 294\ K$

$T_2 \Rightarrow 38\ ^oC + 273 = 311\ K$

$$P_1V_1 = m_1RT_1$$

$$V_1 \Rightarrow \frac{m_1.R_1.T_1}{P_1} = \frac{12\ kg \times (287\ J/kgK).(294\ K)}{(150 \times 10^3)}\ \surd$$

$$= 6{,}75024\ m^3\ \checkmark$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 \Rightarrow \frac{T_2P_1V_1}{T_1P_2} = \frac{150 \times 10^3 (6.75024\ m^3)311\ K}{294K.(250 \times 10^3\ Pa)}\ \surd$$

$= 4{,}2843\ m^3\ \checkmark$

Volume added= $V_1 - V_2$

$= 6{,}7524\ m^3 - 4{,}2843\ m^3$

$= 2{,}46806\ m^3\ \checkmark$

$$m_{added} \Rightarrow \frac{P_2.V_2}{R.T_2} = \frac{(310 \times 10^3\ Pa).(2{,}47806\ m^3)}{(287\ J/kgK).(311\ K)}\ \checkmark$$

$= 8{,}572\ kg\ \checkmark$ (6)

8.3 $V_1 = 0.208\ m^3$ $V_2 = ...$

$t_1 = 30\ ^oC$ $t_2 = 6\ ^oC$

$P = 185\ kPa$

8.3.1 V_2

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_2 = \frac{T_2 V_1}{T_1}$$

$$= \frac{279\ K.(0{,}208\ m^3)}{303\ K} \checkmark$$

$$= 0{,}191\ m^3 \checkmark$$ (2)

8.3.2 V_2

$$\frac{V_1.P_1}{T_1} = \frac{V_2.P_2}{T_2}$$

$$P_2 = \frac{P_1 T_2 V_1}{T_1 V_2} \checkmark$$

$$= \frac{185\ kPa.(279K).(0{,}208\ m^3)}{303\ K.(0{,}089\ m^3)} \checkmark$$

$$= 389{,}113\ kPa \checkmark$$ (3)

[14]

TOTAL: 100

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